VIJAYSINH YUVRAJ BHOSALE

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PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience building Python-based AI applications, Retrieval-Augmented Generation (RAG) pipelines, and machine learning solutions. Experienced in LLM integration, document processing, embeddings, vector search, time-series forecasting, model evaluation, and REST API development. Built end-to-end projects spanning GenAI question answering, predictive maintenance, and data-driven market recommendations, with a focus on practical AI application development.

PROJECTS

Chat with PDF | Retrieval-Augmented Generation (RAG) Application

Technologies: *Python, LangChain, Google Gemini, FAISS, Embeddings, PyPDF2, Streamlit*

- Built a RAG-based document question-answering application that allows users to upload multiple PDFs and ask natural-language questions about their content.
- Implemented a document retrieval pipeline covering PDF text extraction, recursive text chunking, embedding generation, FAISS vector indexing, similarity search, and LLM-generated contextual answers.
- Integrated Google Gemini with Gemini embeddings and FAISS to retrieve relevant document passages and ground responses in uploaded content.
- Developed an interactive Streamlit interface for document upload and conversational querying, demonstrating end-to-end integration of LLM and retrieval components.

Manufacturing Equipment Predictive Maintenance System

Technologies: *Python, Pandas, NumPy, Scikit-learn, Random Forest, FastAPI, Pydantic, PostgreSQL, SQL, Streamlit, SHAP*

- Developed a machine learning application to predict machine failures from sensor data using the AI4I 2020 Predictive Maintenance dataset (10,000 records).
- Built data preprocessing and feature-engineering steps, including temperature-difference and mechanical-power features, for binary machine-failure classification.
- Trained and evaluated a Random Forest classifier, reporting 99% accuracy, 90% precision, 79% recall, and 84% F1-score; validate these figures against the final evaluation results before submission.
- Created a FastAPI REST API for model inference and a Streamlit dashboard for interactive predictions; used SHAP to help interpret feature contributions.

AI-Powered Agricultural Commodity Market Prediction | Crop2Market

Technologies: *Python, Pandas, NumPy, Scikit-learn, TensorFlow, LSTM, XGBoost, Streamlit, QGIS*

- Developed an agricultural price-forecasting and market-advisory application using a dataset of 8,000+ records covering eight agricultural commodities.
- Built an LSTM-based time-series forecasting workflow to predict commodity prices over a seven-day horizon; used XGBoost as a complementary forecasting model, as documented in the project.
- Designed market recommendations that compare expected commodity revenue with estimated distance-based transportation costs to rank selling locations.
- Integrated geospatial distance calculations using QGIS to support market accessibility analysis and data-driven selling recommendations.

TECHNICAL SKILLS

Programming: Python, SQL

Generative AI & LLM Applications: Retrieval-Augmented Generation (RAG), LangChain, Google Gemini, Prompt Engineering, LLM Integration, Text Embeddings, Semantic Search, FAISS, Document Chunking

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Random Forest, LSTM, XGBoost, Time-Series Forecasting, Feature Engineering, Model Evaluation, SHAP

Data & Databases: Pandas, NumPy, Exploratory Data Analysis (EDA), PostgreSQL, MySQL

Application & API Development: FastAPI, REST APIs, Pydantic, Streamlit

Tools: Git, GitHub, Jupyter Notebook, QGIS

EDUCATION

Dr. Babasaheb Ambedkar Technological University, Raigad | B. Tech, Computer Engineering | CGPA: 7.24 | 2022–2026

The New High School and Junior College, Ichalkaranji, Kolhapur | MSBSHSE Class XII | 79.83% | 2020–2022

MAEER's MIT Vishwashanti Gurukul School, Pandharpur, Solapur | CBSE Class X | 85.00% | 2019–2020

LEADERSHIP & ACHIEVEMENTS

ACES — Association of Computer Engineering Students | Technical Coordinator | Jun 2022–Aug 2024

- Coordinated technical workshops and coding events for 300+ students.
- Introduced a communication protocol to reduce inter-team misunderstandings and improve issue resolution.
- Smart India Hackathon 2024 — College-Level Winner | Problem Statement: PSID-SIH1586 | Team: Tech Titans